Revision Report / Response to Reviewers

*Initial Placement for Fruchterman–Reingold Force Model With Coordinate Newton Direction*

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Stephen Kobourov
Technical University Munich
Journal of Graph Algorithms and Applications

Dear Prof. Stephen Kobourov,

Thank you for directing us to the reviewers' comments and for giving us the opportunity to revise our manuscript, and we would like to express our deepest gratitude for your generous and flexible support regarding the situation with the system error.

Based on the reports, we have carefully reviewed all comments and revised the manuscript accordingly. Our responses are provided below in a point-by-point manner. This manuscript has been approved by all authors, and we agree to its submission.

We hope the revised version is now suitable for the JGAA journal and look forward to hearing from you.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda

## A Response to Reviewer A

We thank Reviewer A for the careful and constructive assessment of our manuscript. We have revised the manuscript to address the concerns. The blue text below indicates the reviewer's comments, and our responses are provided in the following paragraphs.

### A.1 Relationship Between the Proposed Approach and the FR Force Model

> First, the authors do not sufficiently consider how well their approach matches the FR force model.
>
> Originally, Fruchterman and Reingold [10] proposed the FR model and solved it by using a pseudo-physical simulation-based algorithm (by following Eades' spring embedder method [8]). By contrast, Kamada and Kawai [21] proposed the KK force model and solved it by a Newton-like method. Also, many succeeding studies followed these approaches, i.e., either by combining simulation-based algorithms with the FR model or by combining Newton-like methods with the KK model. Although Hosobe [19] showed that both models could be solved with a Newton-like method called L-BFGS, the results still suggested that the KK model was more suitable for L-BFGS than the FR model.

**Response:** Thank you for the detailed comment. We agree with these observations.

> However, this submitted paper adopts an unusual approach that combines a Newton-like method with the FR model. I understand that the paper is focused only on the computation of initial layouts, but I think that the validation of adopting this approach is still insufficient.

**Response:** We first acknowledge that several expressions in the previous version of the manuscript may have caused confusion, and we apologize for this.

The key point is that the proposed method is not tied to L-BFGS. Rather, the proposed method is an initial-layout method that can be combined with either the FR algorithm (simulation-based algorithm in the current manuscript), or with L-BFGS. We did not intend to claim that the proposed method should specifically be combined with L-BFGS. However, our previous wording may have given the impression that L-BFGS itself was part of the proposed method. In the previous version, as Reviewer B also noted, we referred to the effectiveness of L-BFGS in the introduction and elsewhere. This may have suggested that L-BFGS was an essential component of the proposed method. However, as illustrated, for example, in Fig. 1, the proposed initialization method and the subsequent optimization method are independent components.

We carefully revised the manuscript to clarify this distinction. If a reader wishes to use the FR algorithm after initialization, they may do so. Indeed, combining the proposed method with the FR algorithm is also within the intended scope of the method. We revised the introduction and related parts of the manuscript to communicate this point more clearly.

Our response continues below.

> It should be noted that the experimental results in Section 5 show that the proposed initial layout algorithm works better when it is combined with the L-BFGS algorithm for computing final layouts. It seems from Figure 6 that the combination of the proposed initial layout algorithm with the FR algorithm for computing final layouts does not work well except in limited cases like jagmesh1.

**Response:** We apologize for the misleading presentation of the results, but we believe that this point was caused by the way the results were presented rather than by the actual numerical results. In Fig. 6, the proposed method outperforms the baseline in almost all cases including jagmesh1, and the differences are sufficiently clear both numerically and visually.

We therefore revised Fig. 6 and its explanation to make the comparison easier to interpret. Specifically, we emphasized the proposed method in bold, renamed several methods for clarity, reordered the columns, and added ruled lines so that the comparison between random initialization and the proposed method is easier to see. We also clarified this point in the main text.

In particular, the proposed method often shows a larger improvement when combined with the FR (simulation-based) algorithm than when combined with L-BFGS. In many cases, the difference between RAND-SIM and CN-SIM is larger than the difference between RAND-LBFGS and CN-LBFGS. In this sense, the effect of the proposed initialization method is often more pronounced when it is combined with the FR (simulation-based) algorithm.

The impression that the proposed initialization method works better when combined with L-BFGS may come from the fact that the final objective values obtained by the FR algorithm are higher than those obtained by L-BFGS. However, this is a comparison between two existing optimization methods and is independent of the proposed initialization method itself. The contribution of the proposed method is that it provides a better initial layout than random initialization.

We hope that the revised presentation and explanation adequately address the reviewer's concern.

> If the authors think that the proposed algorithm should be combined with the L-BFGS, they should adopt the KK model instead of the FR model, which would be more natural and more persuasive.

**Response:** As the explanation above indicates, our position is that the proposed method is effective whether it is combined with the FR (simulation-based) algorithm or with the L-BFGS algorithm. Therefore, we do not argue that it "should be" combined specifically with L-BFGS.

For this reason, we have not adopted the suggestion to replace the FR model with the KK model in the present manuscript. As discussed later, however, we regard combinations with other force models and generalizations in that direction as valuable topics, and we added a separate subsection to discuss them.

At the same time, for readers who simply want to obtain the best final layout, as a general observation based on prior work and independently of the proposed method, that L-BFGS often gives better results than the FR algorithm as an optimization method.

> If the authors think that the proposed algorithm works well for the FR layout algorithm (or another simulation-based algorithm), they should more deeply explore this direction and present clearer results in the paper.

**Response:** To reiterate, our claim is not that the proposed method should be combined with the FR (simulation-based) algorithm or should be combined with L-BFGS. (Indeed, as we later discuss, we added further numerical experiment results for other force models such as HC and Eades, and the proposed method with L-BFGS also shows a clear improvement in those cases. Therefore, the proposed method is not limited to the FR (simulation-based) algorithm or to the FR model.) Rather, both combinations are within the scope of the proposed algorithm. As stated above, we revised the manuscript to make this point clear.

### A.2 Difference Between FR and KK Models

> Second, the authors do not sufficiently consider differences between the FR force model and others, especially the KK model, which becomes apparent especially in Subsection 7.1.
>
> Here the authors first say "Although we utilized the coordinate Newton direction only for the optimization problem (1), we can see that its application is not necessarily limited to the FR force model alone", and then present "objective functions arising from graphs". Does the authors mean that this discussion applies to other force models including the KK model? If so, I think that the authors should adopt the KK model instead of the FR model.

**Response:** Thank you for this insightful comment.

First, as the reviewer suggests, the proposed method is applicable to other force models such as HC and Eades. We added a discussion on the generalization to the HC and Eades layouts in Section 4.5. These force models are also mentioned in Hosobe [19], and the proposed method is indeed applicable to them. We also conducted additional numerical experiments. As described in detail in the revised manuscript, some additional care is required because of differences such as the presence or absence of convexity. Nevertheless, the proposed method was found to be effective in essentially the same way. We added explanations and numerical results on these points to the main text and the Appendix. Since these models are somewhat less widely known than the FR model, we mention them only briefly in the introduction. We also considered changing the title of the paper, but decided to keep the current title.

Second, for the KK model, we believe that the proposed method cannot be applied in the same way as for the FR model. Even if one applies a variant of the proposed method to more general cases, it may not be effective for the KK model. This is closely related to the form of the objective function:

$$\underset{X \in \mathbb{R}^{2\times n}}{\text{minimize}} \quad \Phi_{\mathrm{FR}}(X) = \sum_{(i,j)\in E} \frac{a_{i,j}\|x_i - x_j\|^3}{3k} - \sum_{\substack{(i,j)\in V^2 \\ i<j}} k^2 \log\left(\|x_i - x_j\| + \epsilon^{\mathrm{r}}\right).$$

$$\underset{X \in \mathbb{R}^{2\times n}}{\text{minimize}} \quad \Phi_{\mathrm{HC}}(X) = \sum_{(i,j)\in E} \frac{k_1(\|x_i - x_j\| - a_{ij})^2}{2} + \sum_{\substack{(i,j)\in V^2 \\ i<j}} \frac{k_2}{\|x_i - x_j\| + \epsilon^{\mathrm{r}}},$$

$$\underset{X \in \mathbb{R}^{2\times n}}{\text{minimize}} \quad \Phi_{\mathrm{Eades}}(X) = \sum_{(i,j)\in E} k_1\|x_i - x_j\| \left(\log \frac{\|x_i - x_j\|}{a_{ij}} - 1\right) + \sum_{\substack{(i,j)\in V^2 \\ i<j}} \frac{k_2}{\|x_i - x_j\| + \epsilon^{\mathrm{r}}}.$$

$$\underset{X \in \mathbb{R}^{2\times n}}{\text{minimize}} \quad \Phi_{\mathrm{KK}}(X) \coloneqq \sum_{\substack{(i,j)\in V^2 \\ i<j}} \frac{(\|x_i - x_j\| - l_{ij})^2}{2l_{ij}^2}$$

In the KK model, the main energy terms are defined over all vertex pairs, that is, over sums of the form $\sum_{(i,j)\in V^2\ i<j} \cdot$, rather than over edge pairs of the form $\sum_{(i,j)\in E} \cdot$. This is because the KK energy is defined by first computing all-pairs graph distances and then defining energy terms based on those distances. Therefore, computing forces on a vertex-by-vertex basis does not provide the same advantage as in the FR, HC, and Eades models, and may even lead to slower convergence.

As related work, a method based on another optimization approach called SGD has been proposed for the KK model: J. X. Zheng, S. Pawar, and D. F. M. Goodman, "Graph Drawing by Stochastic Gradient Descent," IEEE Transactions on Visualization and Computer Graphics. In contrast to the proposed method, that approach focuses on pairs rather than vertices when performing optimization, and in that sense it can be regarded as complementary to our work. From the optimization perspective, such SGD-based methods are more suitable for the KK model than the proposed method, and we added a discussion of this point.

Based on these considerations, we removed the previous Section 7.1 and instead revised the manuscript, especially around Section 4.5, to explain these issues more carefully.

> In addition, the authors present the graph isomorphism problem as an application of the proposed algorithm. I think that the KK model would be more suitable for this application than the FR model.

**Response:** Thank you for pointing this out. Regarding the application to graph isomorphism, we agree with the reviewer and recognize that our previous discussion was not sufficiently well motivated. This application is not particularly natural for the FR model and may distract readers from the main contribution of the paper. Therefore, we removed this discussion from the revised manuscript.

> The following might work as hints for the authors' paper revision. Intuitively, the FR model exhibits soft behavior while the KK model exhibits hard behavior. Therefore, the FR model is usually used for dynamic graph layouts like animation and interaction while the KK model is usually used for static layouts. Also, the FR model is sometimes used for very large graphs because simulation-based algorithms for the FR model can be more scalable than Newton-like algorithms for the KK model. One plausible direction for the authors might be to combine the proposed algorithm with a scalable simulation-based algorithm for the FR model and to apply it to very large graphs.

**Response:** We sincerely thank the reviewer for providing this valuable hint. We believe that the soft behavior of the FR model is a very important perspective, especially in dynamic graph drawing scenarios such as animation and interaction. We strongly agree with this viewpoint, and such applications were among the motivations for this study. We added a discussion of this point in Section 6.1, including its relationship to the proposed initial placement method.

### A.3 Notation

> Symbol $E$ is used for two different purposes, one to indicate a set of graph edges and the other to indicate energies between nodes possibly with subscripts and superscripts. It will be better to use different symbols for these different concepts.

**Response:** Thank you for pointing this out. We avoided the symbol conflict by using a dedicated symbol for the edge set, so that the energy terms $U_{i,j}$ are not confused with the edge set.

> I wonder if $(i, j) \in V^2$ with $i \neq j$ in (1) is appropriately intended. This treats both $(i, j)$ and $(j, i)$ in the sum.

**Response:** We originally intended the sum to be over all ordered pairs of distinct vertices, but we understand that this may have caused confusion. We revised the formulation to sum only over unordered pairs of distinct vertices, so that each pair is counted only once.

> Symbol $\epsilon$ appears only once in Subsection 4.4, which seems strange.

**Response:** We clarified the role of $\epsilon$ as the minimum separation in the discrete point set $Q$ and its use in preventing divergence.

> Symbol $f_i$ at line 6 of Algorithm 1 probably lacks the superscript "a".

**Response:** We corrected the missing superscript in Algorithm 1. The update now uses the attractive-only objective $f_i^{\mathrm{a}}$.

> In Subsection 5.3, the word "L-BFGS" that appears as superscripts of $N$ does not seem to be correctly specified. The hyphen mark looks too long.

**Response:** Regarding the reviewer's note on the "L-BFGS" superscripts: the hyphen was already intended as a proper hyphen in math mode, but its appearance could still be misleading. To avoid confusion, we now use the spelling "LBFGS" when it appears inside math expressions, for example in superscripts, while keeping the standard spelling "L-BFGS" in running text.

> In the optimization problem called “objective functions arising from graphs” in Subsection 7.1, “=” probably should be “:=”.

**Response:** Since we removed the previous Section 7.1, this issue no longer appears in the revised manuscript.

> The header of Subsection 8.1 seems to have a problem with the hyphen mark again.

**Response:** We checked the subsection headers and ensured that hyphenation uses proper LaTeX conventions.

# B Response to Reviewer B

We thank Reviewer B for the careful review and constructive suggestions. We have revised the manuscript to improve the framing of the contribution, the discussion of related work, the experimental evaluation, and the organization of the later sections.

## B.1 Related Work on Multilevel Approaches

> In the Related Work section, in 3.1 Multilevel approaches are quickly dismissed. I don’t understand the statement the statement about sfdp being “prior work” to FR - as the latter came way later than the former. This sentence should be clarified.

**Response:** Thank you for pointing this out. We agree that the wording in the previous manuscript was ambiguous.

The source of the ambiguity was the use of the term FR. In the revised manuscript, we distinguish more carefully between the FR force model, which defines attractive and repulsive forces between vertices and corresponds to Problem (1), and the simulation-based algorithm commonly associated with the Fruchterman–Reingold method for seeking an equilibrium of that model.

The paper is concerned with the FR force model itself, as indicated by the title, whereas the simulation-based algorithm is only one possible method for optimizing the corresponding objective. To avoid confusion, we no longer refer to Algorithm 3 simply as the FR algorithm. Instead, we refer to it as a simulation-based algorithm and explain its relation to the classical FR algorithm.

The statement involving `sfdp` was intended to refer to prior algorithmic work on force-directed layout and multilevel optimization, not to the chronological relation between `sfdp` and the FR force model. We have revised the relevant passages to make this distinction explicit and to avoid the misleading implication that `sfdp` predates the FR force model.

## B.2 Discussion of Multilevel Approaches and Potential Applications

> Also, I believe that further space could be given to multi-level approaches, also considering that could be one potential application/extension of this technique beyond this paper.

**Response:** Thank you for the suggestion. As the reviewer notes, we believe that multilevel approaches are a highly promising direction for applying the proposed method. We added a new Section 6.1 to discuss this point.

### B.3 Uncertain Framing of FR Versus L-BFGS

> The framing of this paper is somewhat uncertain. Despite the paper title and general orientation towards FR, L-BFGS is consistently and repeatedly mentioned as a better alternative.

**Response:** Thank you for raising this important point. We have revised the manuscript to clarify that the title refers to the FR force model, namely the optimization problem in Problem (1), and not to a particular algorithm for solving it.

In the previous manuscript, some statements may have suggested that L-BFGS was being presented as an alternative to the FR force model itself. This was not our intention. Rather, L-BFGS and the simulation-based algorithm are two possible methods for optimizing the same FR-based objective. The proposed method is an initial-placement method and is independent of the choice of subsequent optimizer. It can be combined with L-BFGS, with the simulation-based algorithm, or with other optimization procedures.

We have revised the introduction and related passages to make this separation clearer. In particular, we now emphasize that the contribution of the paper is not L-BFGS itself, but an initialization method for problems formulated using the FR force model.

> Even in the related work section, in Section 3.1, only L-BFGS is mentioned as a ready-to-go measure for application in Multilevel approaches.

**Response:** Thank you for pointing this out. We agree that the previous explanation was too narrow. The proposed method can also be used with the simulation-based algorithm, and in some cases its effect may be more visible there because the simulation-based algorithm converges more slowly and is therefore more sensitive to the initial placement.

We have revised the discussion of multilevel approaches to explain that the proposed initialization can be combined with several types of subsequent optimization methods, including simulation-based algorithms, L-BFGS, and potentially other solvers used within multilevel pipelines.

> In the experiments the paper explicitly mentions that FR is consistently outperformed by L-BFGS - then why we should use FR (even with the improved initial placement) at all? The paper does not make a good argument why we should choose that over the other, thus in a way making me question even the title of the paper.

**Response:** We agree that the previous manuscript did not explain this point sufficiently. The purpose of the paper is not to argue that the simulation-based algorithm should always be preferred over L-BFGS. Rather, the paper studies an initialization method for the FR force model, which remains an important graph drawing formulation regardless of which optimizer is used.

L-BFGS often converges faster and achieves better objective values than the simulation-based algorithm in our experiments. However, this comparison concerns the choice of optimizer, not the value of the proposed initialization method. The proposed method can improve the starting point for either optimizer. Therefore, the contribution is best understood as improving the optimization pipeline for FR-based graph drawing, rather than as advocating one particular solver.

We have revised the manuscript, especially the introduction and experimental discussion, to make this framing explicit.

> In general, it's unclear what's the impact of this research and how researchers could build upon it.

**Response:** Summarizing the discussion above, the impact of this paper and the ways in which other researchers can build upon it are as follows.

- Impact: For problems formulated using the FR force model, we propose an initial-placement method that can be effective regardless of whether a simulation-based algorithm or L-BFGS is used afterward, and we demonstrate that it is effective in various cases. Furthermore, in connection with Reviewer A's comments, we also show that a similar idea can be applied to other graph drawing formulations.

- Building upon the work: The proposed method can be combined with other methods for solving problems based on the FR force model, such as multilevel approaches, or with other optimization methods.

We added explanations of these points, especially in the introduction.

### B.4 Expansion of the Evaluation Dataset

> The evaluation yields positive results, but in my opinion it could be further improved. First, I would suggest to expand 5.2 it in terms of number of graphs. More graphs from the Florida library could be used, or another source could be [1]. This expanded datasets already includes 3elt, but expands also on grids, cylinders, and other regular graphs that make up a solid benchmark set, and will provide a wider set of instances to discuss. If not all figures can be included in the main body of the paper, these could be moved in the supplemental material.
>
> [1] Bartel, Gereon, et al. "An experimental evaluation of multilevel layout methods." International Symposium on Graph Drawing. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010.

**Response:** Thank you for the suggestion. As suggested, we expanded the dataset in the Appendix to include additional graphs and substantially revised the presentation of the figures.

For the dataset suggested by the reviewer, we indeed added or created graphs such as `cylinder_30_30`, `gr_30_30`, `sierpinski_06`, and some other graphs. These graphs are "grids, cylinders, and other regular graphs" and are appeared in the paper cited as [1] in Reviewer B's comment or its citing papers. Although the proposed method did not yield significant improvements for all of these graphs, especially when combined with L-BFGS, we observed improvements for the simulation-based algorithm in several cases. We believe that these additional experiments provide a more informative evaluation.

We also attempted to obtain more of the datasets used in the cited paper. Unfortunately, some of the relevant links in the paper [1] appear to be broken, and we were unable to obtain some of the datasets. We also checked the Internet Archive, but the download links were not functional, so we could not retrieve the full dataset. We also experimented with some of the datasets in the paper [1], including parts of the SNAP dataset (`https://snap.stanford.edu/data/index.html`). However, when the number of vertices exceeds ten thousand, the proposed method itself often still works without difficulty, but the subsequent iterations of the simulation-based algorithm or L-BFGS become very expensive and impractical. For such large graphs, combining the proposed method with more scalable techniques, such as multilevel approaches, appears to be a promising direction for future work.

Although we cannot claim to provide a fully comprehensive benchmark set, we report that we expanded the dataset as much as possible to include a variety of graph types.

### B.5 Visualization and Color Map

> Concerning the visualization of the layout, I don't believe that the index colouring adds much. A rainbow map, however, it's not a great choice. If color should be kept, maybe I would consider using a sequential continuous color map with a legend (see [2] for examples).
>
> [2] https://matplotlib.org/stable/users/explain/colors/colormaps.html

**Response:** Thank you for the suggestion. We agree that the previous index-based rainbow coloring did not add much interpretive value and could be visually misleading. We revised the visualizations by replacing the rainbow map with the sequential continuous color map `viridis`, together with an appropriate legend where needed.

### B.6 Effect of Initial Placement Over Time

> One thing that comes out of the charts in Figure 6 is that this different initial positioning has limited effect on the way the objective function decreases over time - which is an interesting phenomenon. In FR, the chart appears shifted below by how "better" the initial positioning is, with the non-CN version never catching up.

**Response:** We believe that this behavior is caused by the slow convergence of the FR algorithm, i.e., the simulation-based algorithm. For the graphs used in Figure 6, there appears to be a nearly unique equilibrium state, and the methods appear to converge toward the same best objective value for each graph. Of course, the initial placement and the subsequent algorithm affect the path through which the objective function decreases. In other words, if sufficiently many iterations are allowed, all methods seem to converge toward the same visually clean graph layout.

Therefore, we believe that it is somewhat inaccurate to say that the difference due to the initial placement remains permanently in the simulation-based algorithm. If a very large number of iterations is allowed, then even for the simulation-based algorithm, as for L-BFGS, the difference due to initialization gradually fades, and the methods eventually converge to the same objective value. However, because the simulation-based algorithm converges slowly, a very large number of iterations would be required, and we stopped the plots at the current range because showing such a long run would make the figures difficult to read.

The discussion continues in the next response.

> Differently, it seems that the objective functions of L-BFGS and CN-L-BFGS tend to converge before the iteration cap. Does this mean that the positive effect of the initial placement tends to fade faster with L-BFGS, despite the better initial condition?

**Response:** The same interpretation applies to this phenomenon. Independently of the merit of the proposed method itself, and simply as a comparison between existing methods, L-BFGS converges faster than the FR algorithm; that is, it approaches an optimal solution more rapidly. Once a method has essentially converged to an optimum, increasing the number of iterations has little effect on the graph layout, because the forces remain nearly balanced. If this optimal state is unique for a given graph, then there is naturally no difference in the final optimum caused by the initial placement. Thus, the effect of the initial placement is more closely related to how quickly the method approaches the optimal state. We revised the manuscript to explain these points clearly.

We also believe it is important to clarify that the objective-function values plotted in the chart do not perfectly correspond to visual quality. For example, the difference in objective value between a completely random state and a moderately clean layout can be much larger than the difference between a moderately clean layout and a fully clean layout. In continuous optimization, when the optimal objective value $\Phi(x^*)$ is known, it is common to plot the gap $\Phi(x) - \Phi(x^*)$ on a logarithmic scale. In our experiments, the optimal objective value is not known, so we simply plot the objective value itself. Therefore, even when the difference in objective value appears sufficiently small—that is, when the difference between RAND-LBFGS and CN-LBFGS appears small—there may still be a sufficiently visible difference in the graph layouts. We added an explanation of these points so that the discussion is more useful and convincing to readers who may have such questions.

### B.7 Singularities in dwt_992 and collins_15NN

> The dwt_992 and collins_15NN singularities could be described more. Would it be possible to generalize, highlighting what makes these two cases so different from each other?

**Response:** Thank you for this suggestion. We agree that the cases of `dwt_992` and `collins_15NN` deserve a more detailed discussion.

We have added a discussion of these two cases in the revised manuscript. In particular, we explain that the main difference between them is not simply whether CN or SA gives a better initial objective value, but whether the initial layout induces a twisted configuration that is difficult for the subsequent optimization algorithm to resolve. In `dwt_992`, the layout initialized by CN contains such a twist, and the subsequent optimization fails to remove it, so the final layout remains twisted. By contrast, in `collins_15NN`, the SA initialization produces a twisted configuration, and the subsequent algorithm requires many iterations to untwist it, whereas the proposed initialization avoids this difficulty.

However, we also found it difficult to formulate a simple general rule that predicts, for an arbitrary graph, whether a given initialization will lead to such a twisted configuration. The final result depends on several interacting factors, including the graph structure, the positional constraints implicitly imposed by the initial layout, the behavior of the subsequent optimization algorithm, and the random seed.

Therefore, instead of claiming a definitive characterization, we revised the manuscript to describe these examples as representative cases showing that the quality of the final layout tends to improve when the initial layout is already close to a desirable final configuration. We hope that this discussion provides a more useful and convincing explanation of the phenomena observed in these two cases.

### B.8 Spike in Objective Function at the Start of Optimization

> Speaking of singularities, in the charts it seems like there is a spike in $f(X)$ right after the 0 seconds mark (like in 3elt, btree or cycle) is that an artifact?

**Response:** Thank you for pointing this out. This phenomenon occurs because, in the initial stage of the FR algorithm, the step size is too large and the solution oscillates. Our original intention was to conduct experiments using exactly the same parameters as external implementations such as NetworkX in order to make the experiments as natural as possible. In that sense, this was intended behavior.

However, we agree that it can easily cause misunderstanding or questions. If properly explained, adjusting the initial step size is also a natural choice for conducting more stable numerical experiments. Therefore, in the revised manuscript, we adjusted the initial step size of the FR algorithm and substantially reduced the oscillation. As a result, we believe that the effect of the proposed method is now easier to see.

### B.9 Organization of Sections 6 and 7

> I am less persuaded from Sections 6 and 7. Section 6 includes considerations that I would include in Section 4, as to further justify your optimization design. I would also simplify the discussion, removing subsections and integrating their contents in Section 4.

**Response:** Thank you for the suggestion. We reconsidered the section organization seriously and revised the manuscript to make the structure clearer. First, we agreed that the material on the KK layout and related topics from Section 6 was more appropriate for Section 4, so we created a new Section 4.5 and moved that discussion there. We believe that this makes it clearer which aspects of the problem the proposed method focuses on. However, moving all of Section 6 into Section 4 would have pushed the experimental results and other material that we wanted to emphasize further toward the end of the paper. For this reason, after consideration, we kept part of Section 6 in place. We also simplified Section 4 by removing some redundant material. Although the revision may not follow the suggestion completely, we believe that it improves the overall quality of the manuscript.

> In Section 7 I would focus on the impact of this approach on graph drawing more specifically. For example, where is the next step for this technique, in the context of graph drawing? Will its presence in a drawing pipeline make a difference, or the gains in quality are not sufficient to justify its use? What about its inclusion in a multi-level pipeline (see also a similar point above)?

**Response:** In connection with Reviewer A's comments, we removed part of Section 7 and moved part of it to Section 4. The previous Section 7.1 gave only a broad suggestion of possible applications. We revised the manuscript so that generalizations to layouts such as HC and Eades are now explained clearly, especially in Section 4.5. We believe that this revision shows more explicitly that the proposed method is also effective for such graph drawing problems, and also explains what properties a problem should have for the current method to be effective. Since we also regard multilevel approaches as one of the most promising applications of the proposed method, we added a new Section 6.1 to discuss this point.

### B.10 Moving the Section 8 to the Appendix

> I would move Section 8 to supplemental material out of the main body of the paper.

**Response:** Thank you. In the revision, we moved the implementation-oriented material (NetworkX implementation details) to the Appendix so that the main narrative focuses on the method, experiments, and discussion.

### B.11 Bibliography

> Bilbiography must be updated. Some papers do not include any venue, others do not have a DOI, others only report their archive version (while journal versions are available).

**Response:** Thank you for the careful note. In the revision, we reviewed the bibliography entries and improved completeness where possible (e.g., adding missing metadata and cleaning local-only fields), and we checked for cases where a journal/conference version should be cited instead of an archive-only version.

## Closing

We again thank the editor and the reviewers for their valuable comments. We believe that the revised manuscript is clearer in scope, better positioned with respect to existing force-directed graph drawing methods, and more explicit about both the strengths and limitations of the proposed approach.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda